Year 12 Maths - Introduction to kinematics

Similar questions - Velocity, rest and total distance

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle moves in a straight line. At time t seconds its acceleration is $a = (6t - 18)$ m s^{-2}. When $t = 0$ its velocity is 24 m s^{-1} and its displacement from the starting point O is 0.

(a) Show that $v = 3t^2 - 18t + 24$. **[2]**

(b) Find the times at which the particle is instantaneously at rest. **[2]**

(c) Find the total distance travelled by the particle in the first 5 seconds. **[5]**

2 A particle P moves in a straight line with velocity $v = 12t - 3t^2$ m s^{-1} for $t \geqslant 0$, starting from a point O.

(a) Find the time at which P next comes to instantaneous rest. **[2]**

(b) Find the maximum velocity of P and when it occurs. **[3]**

(c) Show that after 6 seconds P is back at O, and find the total distance it has travelled. **[4]**

(d) A second particle Q leaves O at the same moment as P and travels in the same direction at a constant 8 m s^{-1}. Find the times at which P and Q are the same distance from O. **[3]**

Question 1

(a) $v = \int (6t - 18)\,\mathrm{d}t = 3t^2 - 18t + c$; $v = 24$ at $t = 0$, so $c = 24$

(b) $3(t-2)(t-4) = 0$, so $t = 2$ and $t = 4$

(c) $s = t^3 - 9t^2 + 24t$. $s(2) = 20$, $s(4) = 16$, $s(5) = 20$. Distance $= 20 + (20 - 16) + (20 - 16) = 28$ m

Question 2

(a) $12t - 3t^2 = 3t(4 - t) = 0$, so $t = 4$

(b) $a = 12 - 6t = 0$ at $t = 2$, where $v = 24 - 12 = 12$ m s^{-1} (a maximum since a changes from positive to negative)

(c) $s = 6t^2 - t^3$. $s(6) = 216 - 216 = 0$. $s(4) = 96 - 64 = 32$, so P goes out 32 m and returns: total distance 64 m

(d) $6t^2 - t^3 = 8t$, so $t(t^2 - 6t + 8) = 0$, $t(t-2)(t-4) = 0$: $t = 2$ and $t = 4$ (and $t = 0$ at the start)